

Classification of human movements by using Kinect sensor

Büşra Açış^{a,*}, Selda Güney^b

^a Department of Electrical and Electronic Engineering, Başkent University, Ankara, Turkey

^b Department of Electrical and Electronic Engineering, Başkent University, Ankara, Turkey

ARTICLE INFO

Keywords:

Long short term memory
Deep learning
Human activity recognition
Machine learning
Kinect sensor

ABSTRACT

In recent years, studies have been carried out to classify human movements in many areas such as health and safety. To classify human movements, image processing methods have also started to be used in recent years. With the help of learning-based algorithms, human posture can be defined in the images obtained by various imaging methods. The predecessor methods of these classification algorithms are machine learning and deep learning. In addition, in recent years, the use of sensors that can detect human joints in perceiving human posture has also increased. The Kinect sensor, developed by Microsoft, is one of the most frequently used sensors because it is not wearable and can detect joints with infrared rays and transfer this information directly to the computer via USB connection. This study used a dataset called CAD60 that included real-time human posture information and images obtained using a Microsoft Kinect sensor, which is available in the literature. This dataset contains data that includes different movements/postures of different people. Within the scope of this study, the performances of these algorithms were obtained by using classification algorithms with the MATLAB program and these performances were compared. The classification algorithms have been used to try to improve the results by using different architectures. When raw data is used, classification accuracy is obtained as 72.60% with one of the machine learning methods, the Cosine K-Nearest Neighbor method. With the feature selection method, this success value has been increased to 74.18%. In addition, when classified by the Support Vector Machines method after the feature extraction process using the Long Short Term Memory method from the deep network architectures, which is the method proposed in this study, the accuracy rate was increased to 98.95%. The best method of classifying human posture was investigated by using different methods and a method was proposed by comparing it with the literature.

1. Introduction

Nowadays, people spend a large part of their day at work. The postures that people perform at work significantly affect the health of the person. For example, if a factory worker performs their work with a hunched posture for a long time, the problem of low back pain may eventually arise, or another example, a software engineer being constantly hunched over the computer will cause muscle spasm. Health problems caused by such posture disorders can occur in countless workplaces such as factories, schools, hospitals, banks. Posture disorders affect people's lives in a negative way. In fact, it can interfere with the functioning of your life as well as at work. For many such reasons, numerous useful studies are being conducted and will continue to be conducted in relation to the correct and effective classification of the posture of people, especially in the field of health [1–3].

Within the scope of this study, it was intended to conduct a study that

classifies human movements by using the data set obtained with the kinect sensor to detect posture disorders shown by people in their workplaces. All studies conducted for this purpose are given below:

1. Classification study was conducted with machine learning methods, and the achievements obtained were tried to be increased by applying the feature extraction method.
2. Using LSTM architecture, feature extraction process was applied on the data set and classification was performed with machine learning methods using the obtained feature matrix.
3. With the transfer learning, classification success was achieved by using different deep learning networks.
4. In order to compare transfer learning with scratch learning methods, classification performance was also measured with a Convolutional Neural Network created from scratch.

* Corresponding author.

E-mail addresses: busraboluk@baskent.edu.tr (B. Açış), seldaguney@baskent.edu.tr (S. Güney).

<https://doi.org/10.1016/j.bspc.2022.104417>

Received 16 September 2022; Received in revised form 4 November 2022; Accepted 20 November 2022

Available online 2 December 2022

1746-8094/© 2022 Elsevier Ltd. All rights reserved.

By comparing the outputs obtained from all sub-studies, the most appropriate classifier for the data set used was tried to be determined.

The study consists of 5 parts in total. In the first part, the purpose of this study is briefly mentioned. In the second part, a literature review was made on the studies conducted in the past in relation to the study performed and the studies of a similar nature and the deficiencies in the literature are given in this section. In the third part, information is given about the methods used in this study. In the fourth part, there is a discussion section and the results obtained at the end of the study. Finally, in the fifth section, there is a conclusion title that summarizes the study and the outputs.

2. Related works

Nowadays, since detecting and defining human activity has a wide range of uses, a lot of studies and innovations have been developed on these topics. One of the most used sensors in the studies on this topic is the Kinect sensor because of how cheap and easy it is to use. S.A. Abdul Shukor et al. conducted a study that detected human falling using a Kinect sensor. The developed system gave the right results under ideal conditions [4].

In the study conducted by Tao Wang et al. a gait analysis was performed for automatic and real-time depression detection using a Kinect sensor. The classification accuracy of the developed model was obtained as 93.75 % [5].

In the study conducted by Baoqi Yao et al., a human activity recognition system using triple Kinect sensors was developed for virtual basketball training. The developed system has been tested with the virtual reality basketball application. According to the results of the experiment, it was observed that the proposed system was usable [6].

In the study conducted by Timeea Sărătean et al., they have developed an autonomous system based on a personal assistant who can monitor a patient receiving physiotherapy treatment at home. According to the results obtained, it was observed that physiotherapists can do physiotherapy exercises remotely [7].

Alexandre de Queiroz et al. conducted a study that applied a real-time approach to gait data obtained with the Kinect sensor for use in clinical trials. According to the results obtained, high accuracy was obtained with knee and hip angles. However, the foot angles adversely affected the operation of the system [8].

The work conducted by Zhenda Xu et al. is in the field of health. In this study, a method that can be used to evaluate spinal deformation has been developed. The method has been experimentally tested. The average RMS value was calculated as 8 mm [9].

In the study conducted by Jyothsna Kondragunta and Gangolf Hirtz, the walking parameters were attempted to be estimated using three-dimensional human posture estimation in the early detection of dementia on the elderly. The results obtained showed that the system was able to work [10].

In the study conducted by Naveen Kumar Mangal and Anil Kumar Tiwari, a filter was proposed to remove noise and improve the quality of the three-dimensional coordinates of the data located around the body to create a movement signature for the kinematic analysis of musculoskeletal disorders. According to the results obtained, it was observed that the range of motion values obtained from the filter proposed in this study increased the success of monitoring the skeletal joint with Kinect [11].

Shalini Nehra and Jagdish Lal Raheja developed a human activity recognition system that is used to monitor and detect the daily activities of a person who is indoors using a Kinect sensor. According to the experimental results, it was observed that the system worked with high accuracy on any data set [12].

In the study conducted by Tan-Hsu Tan et al., a detection system was developed to recognize the daily and abnormal activities of elderly people. Using fourfold cross-validation, the success value of experimental results was assessed by precision, recall, specificity, accuracy,

and F1 score. These success values are respectively 95.5 %, 95.6 %, 99.8, 99.6 % and 95.3 % [13].

In another study, Takuya Kamizono et al. conducted a study on defining the activities of the students to understand the situation of the students while they were in class and to improve the quality of the lesson. K-NN method from ML algorithms was used on the data. The success of the system in recognizing the activities was found to be 0.936 [14].

In the study conducted by Min Li et al., the three-dimensional pose estimation method was tried for babies lying down. According to the results obtained from the method, the average error of the estimated length of the body part was found to be 13.76 mm. In addition, the accuracy of the Percentage of Correctly Localized Parts (PCP) and Percentage of Correct Key Point (PCK) information was observed to be 80.7 % and 86.1 % respectively [15].

In the study conducted by Weyan Ren et al., the Kinect sensor obtained information about the posture of twenty different people lying in bed. The method was tried with SVM architecture in ML method and classification success was found to be 97.1 % [16].

In the study conducted by Ömer Faruk İnce et al., a new biometric system was proposed to detect human activities in three-dimensional space. The K-NN algorithm was used in the ML method for classification and the accuracy of the system was obtained as 86.1 % [17].

In another study, Lucio Tommaso De Paolis and Valerio De Luca tried a method to assess shoulder joint mobility in subjects with different body sizes based on data obtained by the Kinect sensor [18].

In another study conducted by Pramod Kumar Pisharady and Martin Saerbeck presented a multi-class human posture detection and recognition algorithm that does not change with position and scale using geometric properties of the data obtained using Kinect. If the algorithm was tested both by the offline method and real-time application, a classification success rate of 95.78 % was achieved. In addition, the accuracy and recall successes of the model are 100 % and 98.54 %, respectively [19].

In another study, a method was proposed by Wen-June Wang et al. to successfully recognize the different postures of five different people. The classification success of the system was found to be over 99 % [20].

Ahmed Abobakr et al. have presented a method for dealing with a holistic posture analysis that can predict body joint angles from an entry-depth image. The proposed method was observed to achieve an average MAE of 4.67 degrees and an RMSE of 6.64 degrees [21].

In another study, Ravindu Kumar et al. presented a method related to the dual Kinect system that increases the range of gait analysis capture. In the experimental results of the method, 4 % error was obtained in length measurements and 5 degrees error in flexion measurements [22].

The study by Yifeng Xu et al. proposed an advanced feedforward artificial neural network that recognizes three target exposures on the dataset obtained using the Kinect sensor. As a result of real-time experiments, the accuracy of the system in fall detection was found to be 98.5 % [23].

Heng Sun et al. have developed software for detecting and stimulating students' sitting postures in real-time via the Kinect sensor. Experimental results have shown that the system is feasible [24].

Shehzan Haider et al. have proposed a method to control rehabilitation exercises to be performed in the postoperative healing of musculoskeletal disorders. According to the results obtained, HF-LSTM method is error-free, while GCN-LSTM works with RMSE loss of 0.191 [25].

In a study conducted by Dinesh Kumar Vishwakarma and Konark Jain, they proposed a system of human movement recognition using the concept of dimensional transformation. According to the results obtained; a success rate of 94.13 % was achieved in MSR Action3D dataset; 93.34 % in the Berkeley MHAD dataset; 95.70 % in TST Fall Detection Data Set and 86.84 % in NTU-RGB + D [26].

In another study, Ahmed Ferdous et al. presented a method for automatically recognizing emotion from body movements. The obtained

classification results were obtained as 90 %, 96 % and 86.66 % respectively [27].

In the study conducted by Jaeyong Sung et al., human activity was detected through RGBD images obtained with the Kinect sensor. Environments where daily activities are carried out are defined as structured and unstructured environments. An unstructured environment is defined as a messy home or office. A structured environment is the opposite. The dispersion of the environment makes it difficult to detect people's movements. For this reason, in this study, the detection of the movements was carried out in an unstructured environment. The algorithm performed in this study was carried out according to a hierarchical MEMM consisting of a series of sub-activities of a person's activity. Since it was aimed to compare the achievements of the algorithm in different environments, activity recognition tests were performed in five different environments. Algorithm success was evaluated by precision and recall criteria. In addition, two different success results were obtained as previously defined in the algorithm and never defined (a new person). As a result of the algorithm, detection and classification results were obtained with 84.7/83.2 % and 67.9/55.5 % precision/recall measurement in "seen" and "new person" situations respectively [28].

Prabal Datta Barua et al. have developed a new method to accurately detect COVID-19 disease in their study. In this study, nine deep feature extraction methods were applied by creating hybrid deep features using three deep networks (AlexNet VGG16 and VGG19) together. The maximum classification success of the proposed model with the SVM classifier was found to be 99.64 % [29].

Haixia Liu et al. presented a new method for detecting breast cancer. The method introduced a new feature generation technique called the grid-based deep feature generator. With the proposed method, the error values of the entire deep feature generator can be calculated. Also, the three best feature vectors are chosen as the final feature vectors. The classification accuracy of the model classified by the deep neural network was obtained as 97.18 % [30].

In another study, Prabal Datta Barua et al. presented a method that can automatically detect retinal disorders with the transfer learning method. In the method, first feature extraction was applied. Then, feature selection was applied with ReliefF and classification was made using a second-order support vector machine (QSVM). The method has been tested with two different OCT datasets (DB1 and DB2). The classification accuracies obtained for the DB1 and DB2 datasets were 97.40 % and 100 %, respectively [31].

Emrah Aydemir et al. carried out a study that classifies whether the mask used during epidemic periods is used correctly or not. The proposed method in the study consists of three stages. Firstly, 1000 features were extracted from each image using ResNet101 and DenseNet201 network architectures. Then, the Relief selector was used to identify the features with high discrimination, and finally the system was tested with the support vector machine classifier. Method: It has been tested for 3 different situations as masked, unmasked and incorrect mask use. The classification accuracy obtained for these cases was 95.95 %, 97.49 % and 100.0 %, respectively [32].

When all these studies in the literature are examined, it is observed that the success of motion classification decreases with the increase in the number of classes. It has been observed that new studies are required to increase the classification success in multi-class problems. In this study, it was aimed to increase the classification success for 13 different classes.

In this study, the classification of human movements was made utilizing the data set used in the study of Jaeyong Sung et al. [28]. However, all associated studies show that success decreases as the number of activities to be classified increases. Studies are needed to increase the success of this classification. In addition to the studies in the literature, it is also aimed to detect the posture disorders of the employees in all kinds of work environments such as factories, schools, hospitals. For this purpose, machine learning methods, feature extraction with LSTM, learning transfer and scratch learning methods were used to obtain the



Fig. 1. Kinect sensor [35].

most successful classification result and the most appropriate method was proposed.

3. Materials and methods

3.1. Kinect sensor and dataset

The first generation, Kinect v1 (Fig. 1), is a set of motion detection input devices first released by Microsoft in 2010 and developed for game consoles and Microsoft Windows PCs [33]. Later, Microsoft further developed Kinect v1 and released Kinect v2 in 2014 [34].

The Kinect Sensor consists of a color camera, an Infrared (IR) emitter, an IR depth sensor, an engine for tilting, a microphone array, and an LED light [36]. Three types of information can be received with Kinect devices: color images, 3D depth images, and bone information corresponding to the 3D imaging field [34,37]. Fig. 2.a shows the RGB image obtained with the Kinect sensor and Fig. 2.b shows the depth image obtained with the Kinect sensor.

In this study, it is aimed to compare the success values of different algorithms with each other in the classification of human postures. Kinect sensors are often preferred for the detection of human posture due to their advantages such as not being wearable and costly. In this study, a ready-made data set known as CAD60 is used, which includes human postures obtained with the Kinect sensor, used in the study of Jaeyong Sung et al. [28]. Fig. 3 shows some sample frames from the data set. While obtaining the data set, sensor data were obtained in five different environments including office, kitchen, bedroom, bathroom and living room. The data set was created with four people, two men and two women.

The number of image data obtained from four different people is as follows; 17,918, 16,805, 17,577 and 20,360. In total, studies were carried out on 72,660 data. In this study, classification studies were applied with 2 different data sets and comparison was made in terms of data. The first of these is the RGB images obtained through the Kinect sensor. The other is the joint data obtained by the Prime Sensor NITE 1.3 algorithm, which monitors 15 body joints. Fig. 4 shows the joints used in the data set. Joint data include joint orientation and/or joint position information. The joint orientation consists of a matrix of 3x3. Joint position information is obtained with 3 variables as x, y and z. There are 11 joints with both joint orientation and joint position. These are; head, neck, torso, left shoulder, left elbow, right shoulder, right elbow, left hip, right hip. Apart from this, there are only 4 joints with a joint position. These are; left hand, right hand, left foot and right foot.

There are 13 labels in the data set. The label values are sorted from 1 to 13 to represent each movement. These tag values and their equivalents are given in Table 1.

There are some limitations regarding the data set. These:

- In the dataset, the background is fixed and cluttered.
- The movements are of medium complexity.
- The complexity level variation between movements is moderate.
- Variety in execution style is low between different topics or repetitions.



Fig. 2. Kinect sensor RGB image and depth image [28].



Fig. 3. Images related to data set [28].

3.2. Methods

Four different methods were used in this study. The flowchart of the methods used is given in Fig. 5.

According to Fig. 5, Sensor raw data is analyzed in two parts: sensor data and RGB image data. Three different methods were applied with the sensor data. First, the success of the raw data was achieved by machine learning methods without applying any operation on the sensor data. In the second stage, backward feature selection method was applied and machine learning methods were used for classification. Finally, feature selection was performed using the LSTM method. By using the obtained feature matrices, classification process was carried out with machine learning methods. Two different applications were carried out with RGB image data. First, classification success was achieved with the learning from scratch method. In the second stage, feature extraction was performed using 3 different architectures in the learning transfer method. And then machine learning methods were used for the classification of the extracted features.

The problems to be solved with the approaches in machine learning (ML) are classified as supervised learning or unsupervised learning.

Clustering is given as an example of unsupervised learning while classification and regression are examples of supervised learning [38]. In this study, classification will be made, and machine learning algorithms are examined for this stage.

There are many methods developed for classification in machine learning. The Decision Tree, Linear Discriminant Analysis, Support Vector Machine (SVM), Nearest Neighbor Classifier (K-NN) and Ensemble methods are the methods examined within the scope of this study [39–43].

Operations performed on large data sets take a lot of time. In addition, not all of the features in the dataset may be desirable to be used in the work that is performed, or some features may be suppressing other features that are present during operations, which as a result might be affecting success. For this reason, features selection processes can be performed to reduce the algorithm's uptime, increase the prediction performance, and present a better understanding of the data in machine learning applications [44–46].

In this study, the features obtained from sensor data were examined and their effects on classification success when feature selection was applied were examined. The Backward Feature Selection (BFS) method

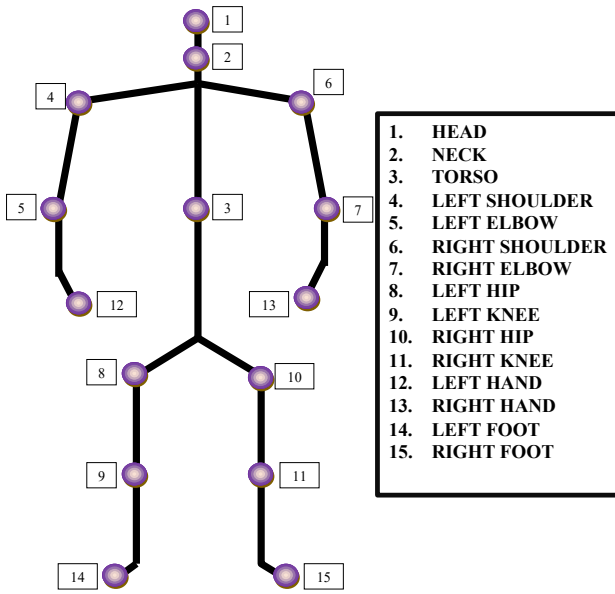


Fig. 4. Joints Used in the Data Set.

Table 1
Label information.

Label Number	Label's Movement Equivalent
1	Call
2	Writing on the Whiteboard
3	Drinking Water
4	Gargling with Water
5	Brushing Teeth
6	Wearing Lenses
7	Talking in a Chair
8	Relaxing in a Chair
9	Cooking (Chopping)
10	Cooking (Mixing)
11	Opening a Medicine Box
12	Working on the Computer
13	Standing Still

was used as the feature selection method. The main purpose of BFS is to create the most efficient training set in terms of time and success from the data set. In this method, the feature extraction process is in the way that the features are processed from many-to-few, backward [47–50].

In this study, Deep Learning method was used for image-based classification of human movements. Deep learning is a machine learning technique that learns from samples that are made up of successive layers and more efficient representations can be obtained when each layer is processed [51]. CNN is a type of neural network developed under the deep learning architecture and has been widely used in the field of classification in recent years [52]. CNN architecture is mostly used to classify and define image data; used in object detection and object tracking systems [53].

The CNN method generally consists of two main parts, the feature extraction and classification section [54]. The feature extraction part consists of one or several convolution layers and pooling layers [55]. For CNN, the layers are generally listed as convolution layer, pool layer, and fully connected layer [56,57]. The convolution layer is the main layer used in the CNN architecture to perform feature inference. In the pooling layer, the number of parameters decreases. The data generated by the layers before it, is compiled to create classification information in the fully connected layer [58].

In the CNN architecture, classification is performed using RGB images. In the learning from scratch method, RGB images were classified

by the CNN model, which consists of 7 layers [41]. Another method is the use of transfer learning methods. The architectures used for this are: DenseNet-201, Xception, and ResNet-50 [59–61].

The LSTM network was developed as an RNN variant by Hochreiter et al. in 1991 [62]. LSTM architecture enables learning of long-term dependencies on sequence data such as signal, sound, text, and other time series. LSTM is an algorithm that can understand which data should be stored for a long time, which data should be discarded, or which data should be read [63].

4. Result

The data set used in this study includes data obtained from four different people. These people are numbered 1, 2, 3 and 4. Learning-based methods require training and test sets during the build and testing phase of the model. In order to make this distinction in the most efficient way in the data set used, a training set was created with the data of 3 people in the data set with different trials, and a test set was created with the data of the remaining person. For example; While the training set was created with the data of people numbered 1, 2, 3, a test set was created with the data of person number 4. Training and test sets are divided into 4 different ways in total. With these distinctions, classification successes have been achieved by using different methods based on machine learning. By examining the results obtained for all data set separations, the classifiers with maximum accuracy were determined. According to the results obtained two of the training and test set distinctions gave maximum success in the Subspace Discriminant method. The other two maximum success values came from the Cosine KNN and Linear SVM methods. In Table 2, the classification achievements obtained by the methods given above for each training and test distinction are given. The average success value was obtained for each method. As a result, the highest average success was achieved with the Cosine KNN method in machine learning method with 72.60 %.

When Table 2 is examined, the lowest classification accuracy was obtained in training set:124 and test set:3. Three people in the data set (person 1, person 2, and person 4) are right-handed. However, person 3 is left-handed. In this data set separation, training was done with the right joints. However, a low success rate was observed as the testing phase was performed with the left joints. Looking at the standard deviations, the lowest value was obtained in the Linear SVM method. It is understood from this that there is less variation in the successes for data set separations in this classifier.

The maximum classification success obtained with the raw data was below the success value of the study in which the data set was obtained. The reason for this is that there are data in the raw data set that will adversely affect success or there are data that are unrelated from each other. This also increases the processing load. For this purpose, feature extraction was implemented to get rid of unrelated data, increase the accuracy result and reduce the processing load.

While applying backward feature selection, reference data set separation was selected and operations were performed on it. For this, the results obtained with the raw data set results were examined and the data set separation with the highest accuracy was determined as training set:134 and test set:2. Backward feature extraction was applied using this dataset separation. The accuracy values obtained when each feature is subtracted for the training set:134 and test set:2 state is given in Fig. 6.

While the raw data success for training set:134 and test set:2 is 91.6 %, the success for the last remaining attribute (left elbow) is 56.5 %. In the backward feature selection process, the classification success first increased with decreasing number of features and then started to decrease. Classification success peaked when 5 features were removed (94.2 %). This graph structure is defined as the peaking phenomenon [64,65]. In addition, 5 features extracted for the peak phenomenon are defined as the optimal number of features [65]. The five extracted features, in order; right knee, head, left hip, left hand and right hand. Since head movement and right knee position were not a determinant feature

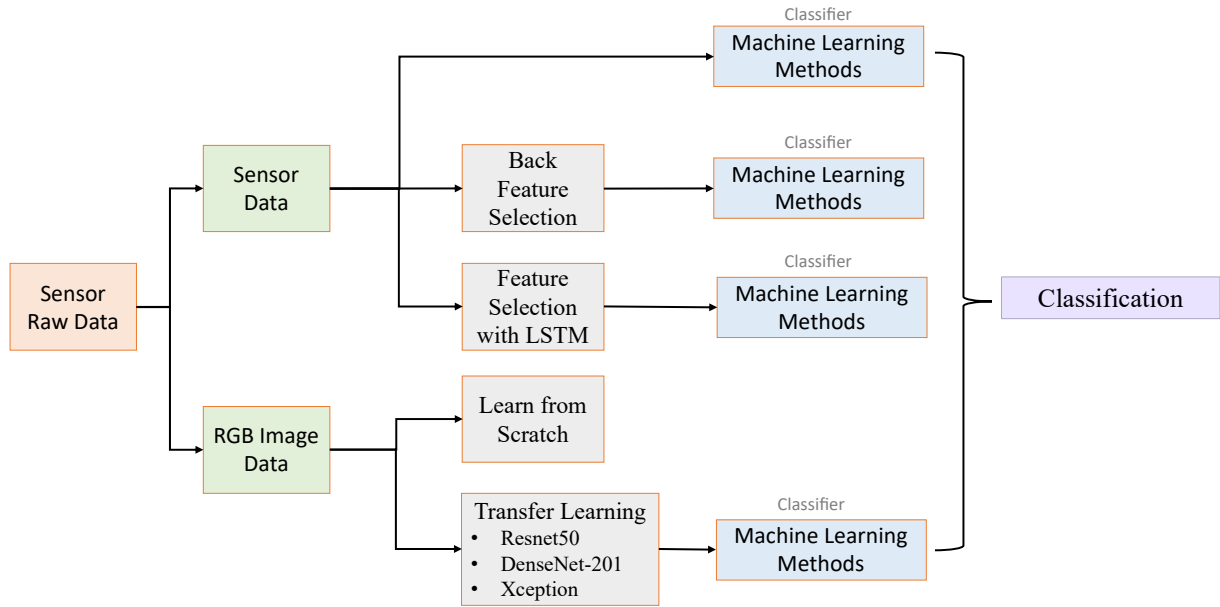


Fig. 5. Methods used in the study.

Table 2

Success values for the training and test dataset.

Data Set	Subspace Discriminant	Linear SVM	Cosine KNN
Training Set: 123 Test Set: 4	%81.1	%64.3	%74.5
Training Set: 124 Test Set: 3	%49.8	%54.8	%54.6
Training Set: 134 Test Set: 2	%76.7	%79.4	%91.6
Training Set: 234 Test Set: 1	%74.1	%63.1	%69.7
Average Success	%70.35	%65.40	%72.60
Standard Deviation	14.05	10.24	15.24
Parameter	Ensemble method: Subspace Learner type: Discriminant Number of learners: 30 Subspace dimension: 72	Kernel function: Linear Kernel scale: Automatic Box constraint level: 1 Multiclass method: One-vs-One Standardize data: true	Number of neighbors: 10 Distance metric: Cosine Distance weight: Equal Standardize data: true

for movements found in the data set, removal of these joints increased success. On label number 2 of the data set, subjects write on a whiteboard. While three of the subjects were right-handed, one of them performed the writing process using their left hand. As can be seen from this, this dataset has less effect on the data set of left joints for this separation. This may be considered for the third joint removed. The last two remaining joints in the feature extraction process are the right elbow and the left elbow. Elbows play a significant role in performing many manual activities. In other words, the elbows direct movement. For this reason, these two joints are left to the last.

The features determined for the reference data set separation were extracted from the other data set separations in the same order. Then, classification accuracies were obtained by using machine learning methods for all data set separations. Results for different training and test dataset distinctions are given in Table 3.

In the BFS method, as in the previous method, the classifiers in which the maximum successes of all data set separations were obtained were

determined. The success values obtained for all data set separations for these classifiers are given in Table 3. The classifiers are the same as the classifiers obtained in the previous step. When Table 3 is examined, it is observed that there is a slight increase in the classification successes obtained for data set separations. In addition, the average success value was taken for each method. When the results obtained are examined, the average success value obtained in the Cosine KNN method increased from 72.60 % to 74.18 % with BFS. However, in the BFS method, the average success rate was higher with the Subspace Discriminant method (74,88 %). Standard deviations have increased compared to the previous stage. The lowest value was obtained in the Linear SVM method, as in the previous stage.

In another sub-study, feature extraction was applied using LSTM architecture. At the end of this process, training and test sets were created using the resulting feature matrix. Machine learning-based methods were used for the classification process. This method has been tested in four different ways, as in the machine learning method. Two of the training and test set distinctions gave maximum success in the Logistic Regression Kernel method. The other two maximum success values were obtained in the Medium Neural Network and SVM Kernel methods. The average success value was obtained for each method. Table 4 shows the classification achievements obtained by the methods given above for each training and test set distinction.

When the average success and standard deviation results given in Table 4 are examined, it is seen that there is a great improvement compared to the results obtained in the previous methods. It is understood from the values that there is not much change in the classification accuracies obtained in the data set separations. All classification accuracies achieved are above 90 %.

The maximum average classification success achieved as a result of feature extraction with LSTM architecture was found to be 98.85 %. When a feature was extracted with LSTM, more features were obtained than in the raw data set. At the same time, LSTM is a method of establishing relationships between features. Therefore, it is used frequently used in time series problem [66,67]. Here, too, the reason for its high success is thought to be the correlation between the sensor data obtained from the joints. In addition, the increase in the number of features increased the classification success of the data set. Additionally, the SVM method is a robust method when the number of features is high and works with high accuracy [68]. Fig. 7, the confusion matrix is given for the result obtained for all dataset separations.

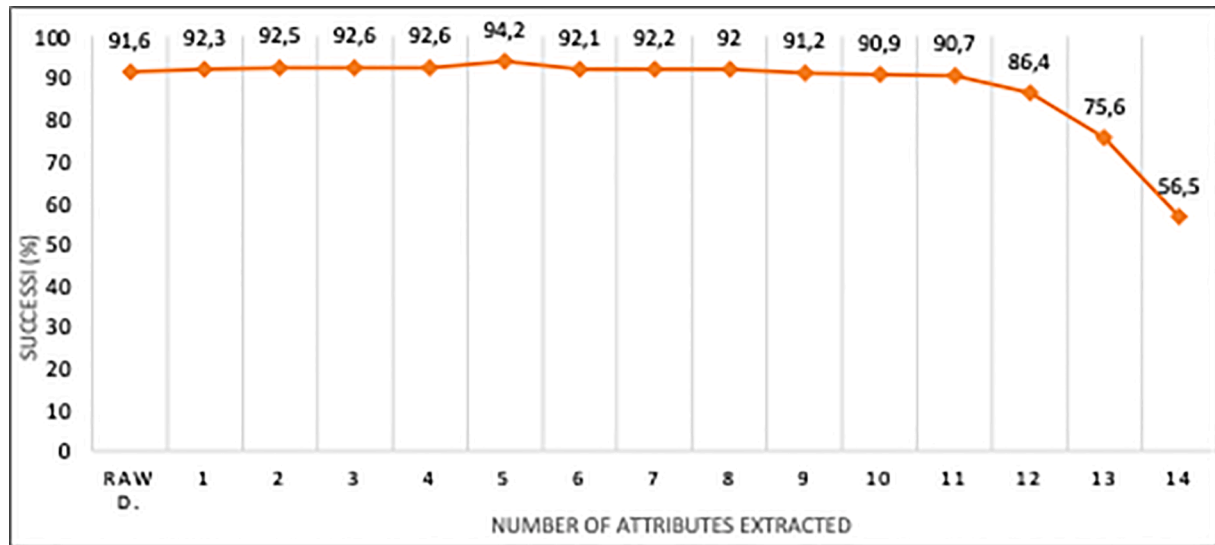


Fig. 6. Classification achievement after BFS process.

Table 3

Achievement values for training and test data set as a result of BFS.

Data Set	Subspace Discriminant	Linear SVM	Cosine KNN
Training Set: 123 Test Set: 4	%82.7	%65.7	%76.2
Training Set: 124 Test Set: 3	%48.9	%51.2	%55.7
Training Set: 134 Test Set: 2	%90.1	%82.8	%94.2
Training Set: 234 Test Set: 1	%77.8	%63.7	%70.6
Average Success	%74.88	%65.85	%74.18
Standard Deviation	18.04	12.99	15.91
Parameters	Ensemble method: Subspace Learner type: Discriminant Number of learners: 30 Subspace dimension: 51	Kernel function: Linear Kernel scale: Automatic Box constraint level: 1 Multiclass method: One-vs-One Standardize data: true	Number of neighbors: 10 Distance metric: Cosine Distance weight: Equal Standardize data: true

The confusion matrices obtained for maximum success values in the dataset separations are given above. When the figures are examined in order, firstly, the confusion matrix obtained for the separation of the training set:123 and test set:4 data sets is seen. The maximum classification success achieved for this dataset separation is 97.9 %. According to the resulting confusion matrix, erroneous results were obtained for a total of 5 moves. Label values determined for these movements are 3, 4, 5, 6 and 10. The maximum success value obtained for the training set:124 and test set:3 data set separation is 98.8 %, and 7 movements in this data set separation are incorrectly (numbered 1, 3, 4, 10, 11, 12 and 13). tags) are classified. The maximum success value obtained in the separation of training set:134 and test set:2 data sets is 99.3 %. There was an error in classifying only one movement in this dataset separation (label 2 – writing on the whiteboard). The maximum success achieved in the last data set separation was misclassification of movements with labels 4, 9 and 11, and 99.8 % classification accuracy was obtained in this data set separation. In general, there was no misclassification in the activities of talking in a chair (label 7) and resting in a chair (label 8) activities. Of the 13 activities, only these two activities were performed sitting down. For this reason, it is thought that the success of classification is high by distinguishing it more clearly from other movements.

The second analysis within the scope of this study is the classification of human movements according to the RGB image obtained from Kinect sensors. For this purpose, the classification process was performed using

Table 4

The accuracy results of machine learning algorithms with features extracted from LSTM.

Data Set	Medium Neural Network	Logistic Regression Kernel	SVM Kernel
Training Set: 123 Test Set: 4	%97.7	%97.9	%97.8
Training Set: 124 Test Set: 3	%98.8	%98.8	%98.6
Training Set: 134 Test Set: 2	%97.8	%98.7	%99.3
Training Set: 234 Test Set: 1	%90.7	%99.8	%99.7
Average Success	%96.25	%98.80	%98.85
Standard Deviation	3.23	0.79	0.83
Parameters	Number of fully connected layers: 1 First layer size: 25 Activation: ReLU Iteration limit: 1000 Regularization strength (Lambda): 0 Standardize data: Yes	Learner: Logistic Regression Number of expansion dimensions: AutoRegularization strength (Lambda): Auto Kernel scale: Auto Multiclass method: One-vs-One Iteration limit: 1000	Learner: SVM Number of expansion dimensions: AutoRegularization strength (Lambda): Auto Kernel scale: Auto Multiclass method: One-vs-One Iteration limit: 1000

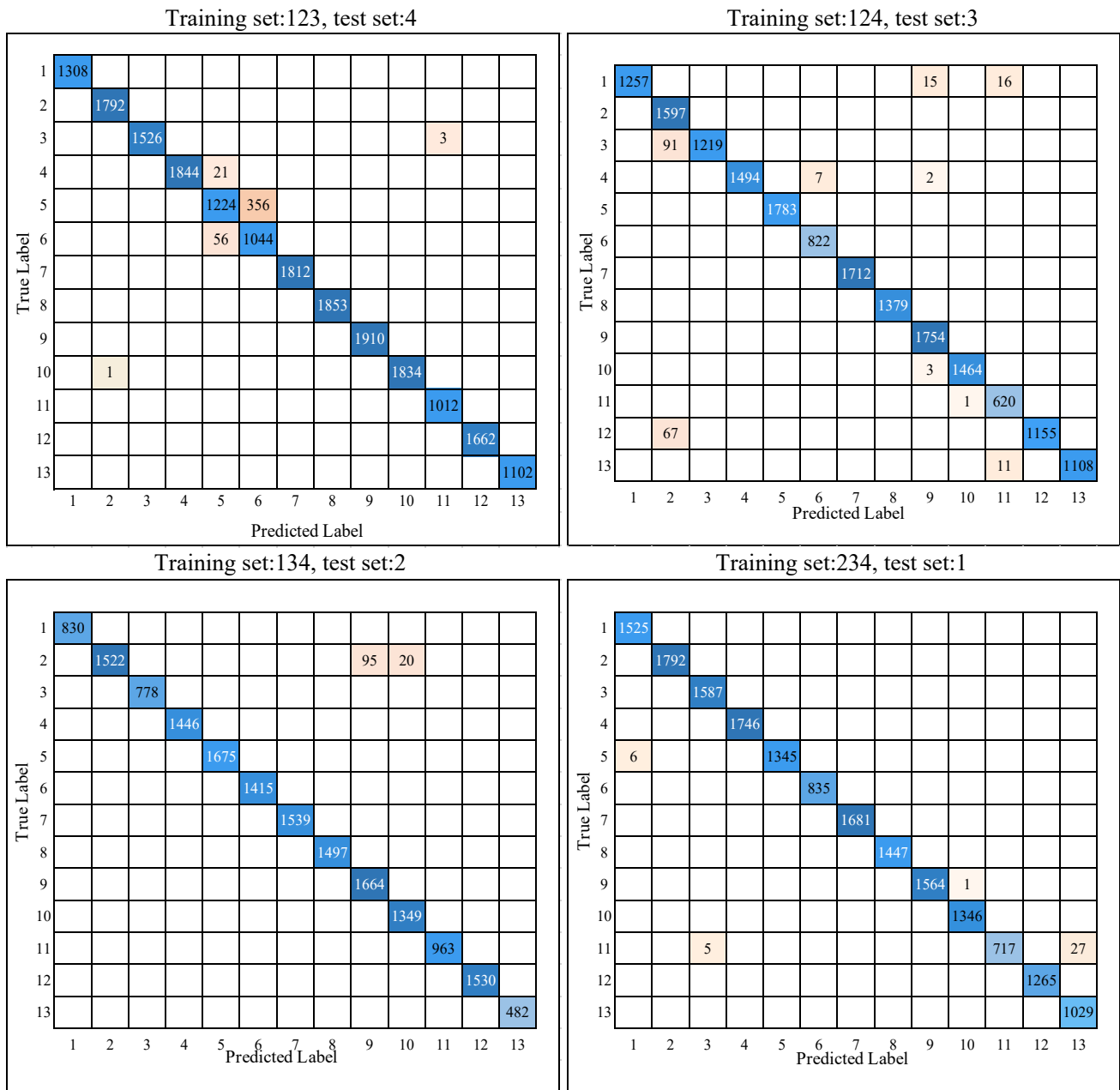


Fig. 7. Confusion Matrix for features extracted from LSTM.

Table 5

Success values obtained with learning from scratch.

Data Set	Success (%)
Training Set: 123 Test Set: 4	%23.51
Training Set: 124 Test Set: 3	%19.71
Training Set: 134 Test Set: 2	%28.12
Training Set: 234 Test Set: 1	%25.61
Average Success	%24.24
Standard Deviation	3.56

72,660 RGB images in the CNN architecture, which is the most used in image processing. Training and testing data were created by separating them as in the previous stages. The data set is classified by the CNN model, which consists of 7 layers with learning from scratch. Table 5 shows the classification success achieved for each training and test data set distinction. The tiers used for the resulting classification success are

listed below:

- Image input layer,
- Convolution layer,
- Relu layer,
- Max pooling layer,
- Fully connected layer,
- Softmax layer,
- Classification layer.

The average success value obtained in the learning from scratch method is 24.24 %. The success value of this method has been exceptionally low compared to other methods. There are thought to be two main reasons for this. First, the learning from scratch method requires exceptionally large data sets to train the network in depth. However, the network could not be adequately trained due to the lack of large volumes of data in the dataset used. As a result, the lack of sufficient distinguishing features resulted in a low achievement value. Another reason for the decrease in success is the difficulty of adjusting the

Table 6

Maximum classification achievements by Resnet50 network architecture.

Data Set	Subspace Discriminant
Training Set: 123 Test Set: 4	%63.6
Training Set: 124 Test Set: 3	%54.9
Training Set: 134 Test Set: 2	%59.6
Training Set: 234 Test Set: 1	%65.9
Average Success	%61.0
Standard Deviation	4.83
Parameter	Ensemble method: Subspace Learner type: Discriminant Number of learners: 30 Subspace dimension: 500

Table 7

Maximum classification achievements by DenseNet-201 network architecture.

Data Set	Linear SVM	Cubic SVM	Medium Neural Network	Narrow Neural Network
Training Set: 123 Test Set: 4	%47.1	%46.9	%41.9	%30.4
Training Set: 124 Test Set: 3	%41.8	%45.6	%56.6	%46.5
Training Set: 134 Test Set: 2	%63.6	%67	%60.7	%54.3
Training Set: 234 Test Set: 1	%57.6	%60.5	%66.7	%73.1
Average Success	%52.53	%55	%56.48	%51.08
Standard Deviation	9.88	10.46	10.56	17.74
Parameter	Kernel function: Linear Kernel scale: Automatic Box constraint level: 1 Multiclass method: 1-vs-1	Kernel function: Cubic Kernel scale: Automatic Box constraint level: 1 Multiclass method: 1-vs-1	Number of FCL: 1 First layer size: 25 Activation: ReLU Iteration limit: 1000 Regularization strength: 0	Number of FCL: 1 First layer size: 10 Activation: ReLU Iteration limit: 1000 Regularization strength: 0

number of layers and parameter values. In order to establish a deep network structure that learns from scratch, it is necessary to determine the appropriate layers for the data set. However, this task is both very

Table 8

Maximum classification achievements by Xception network architecture.

Data Set	Narrow Neural Network	Wide Neural Network	Bilayered Neural Network	Cosine KNN
Training Set: 123 Test Set: 4	%58.1	%71.9	%47.3	%60.5
Training Set: 124 Test Set: 3	%56.7	%58.9	%66.2	%54.7
Training Set: 134 Test Set: 2	%43.2	%49.4	%48.5	%61.3
Training Set: 234 Test Set: 1	%72.8	%66.1	%50.3	%49.1
Average Success	%57.70	%61.58	%53.08	%56.40
Standard Deviation	12.10	9.70	8.84	5.69
Parameter	Number of fully connected layers: 1 First layer size: 10 Activation: ReLU Iteration limit: 1000 Regularization strength: 0	Number of FCL: 1 First layer size: 100 Activation: ReLU Iteration limit: 1000 Regularization strength: 0	Number of FCL: 2 First layer size: 10 Second layer size: 10 Activation: ReLU Iteration limit: 1000 Regularization strength: 0	Number of neighbors: 10 Distance metric: Cosine Distance weight: Equal Standardize data: true

costly and very time-consuming. In addition, these layers contain many parameters that need to be fine-tuned. For this reason, it is a particularly challenging task to optimize all the parameters of the network created. These situations adversely affect the success of the system. This leads us to transfer learning. In this method, feature extraction is implemented using the Resnet50, DenseNet-201, and Xception network architectures. When using the Resnet50 network architecture, the maximum success value was achieved with the Subspace Discriminant classifier. Table 6 shows the classification successes for all dataset distinctions.

With the Resnet50 network architecture, feature extraction was applied to all data set separations. Then machine learning method was used for classification. Maximum classification success was achieved in the Subspace Discriminant method in all data set separations. The average success achieved was found to be 61 %. In the second stage, DenseNet-201 network architecture was used for feature extraction. The resulting feature matrices were classified using machine learning methods. This process was applied sequentially in all data set separations and the maximum classification success for each was found in a different classifier. The classification successes and average success values obtained for these classifiers of data set separations are given in Table 7.

When the results are examined, the classification successes obtained with the DenseNet201 network architecture are lower than the ResNet50 network architecture. The maximum average success value obtained was 56.48 % with the Intermediate Neural Network classifier. Finally, the network architecture used for feature extraction was Xception. All data set separations were classified by machine learning method as in the previous stages. Each data set separation gave the maximum success value in a different classifier. The results obtained are given in Table 8.

The maximum average success value obtained by using Xception network architecture was found to be 61.58 % in the Wide Neural Network classifier. The maximum success value of the transfer learning method was achieved in this network architecture. According to the learning from scratch method, the success value has increased, because the network architectures used here are models that are trained with large data clusters and can generalize well. However, the network architectures used here are also not suitable for the data set used in this study. Extracting the features has not been able to yield complete results. In addition, with these network architectures, work is usually done for image recognition.

5. Discussion

The data and RGB images in the data set have been carefully examined and labeled. The dataset contains 13 class information containing human movements. The data set contains data obtained from four different people. In order to separate the data set as training and test data, the training set was created with the data of 3 people and the test set was created with the data of the remaining person, and classification success was achieved. In the first stage, in order to examine the

Table 9
Compare of overall classification achievements.

Methods	Success
Sung J. et al. [28]	%84.70
Raw Data	%72.60
BFS	%74.88
Feature extraction with LSTM	%98.85
Learning from scratch	%24.24
Transfer Learning	%61.58

classification success of the raw data set, a maximum classification success was obtained in each data set separation and a detailed analysis was performed by averaging the results. As a result, the maximum average classification success obtained was found to be 72.60 % in the Cosine KNN classifier.

In machine learning, feature selection method has been applied to increase classification success. At this stage, backward feature selection method was used. There are a total of 15 attributes representing the joints in the dataset. On the data set separation (training set: 134 and test set: 2), where the maximum classification success was achieved on the raw data set, the features were extracted sequentially and maximum success was tried to be achieved. This process continued until only one attribute remained in the data set. In practice, the maximum success is obtained when 5 features are removed. In this case, classification success was achieved for the same 5 features extracted on other data set separations. The maximum average success value obtained as a result of this was found to be 74.88 % in the Subspace Discriminant classifier.

In the next step, feature extraction was applied using LSTM. After the feature extraction was applied, the classification success of the system was obtained by using 29 different classifiers based on machine learning. This process was performed sequentially for each data set separation and the maximum average success value was obtained as 98.85 % in the SVM classifier.

In the CNN architecture, classification is performed using RGB images. At this stage, RGB images were classified with the CNN model consisting of 7 layers by learning from scratch. Classification success was obtained for each data set separation and the average success value was found to be 24.24 %.

In the last stage, the learning transfer method was used. In this method, feature extraction was applied using Resnet50, DenseNet-201 and Xception network architectures and the classification success of the system was obtained by using 22 different classifiers. At the end of

Table 10
Pseudo-code for the feature extraction with LSTM algorithm.

1. Input: XTrain and XTest data from source
2. Initialize features, classes3. Initialize the LSTM layers and define parameter values
3. Initialize the LSTM layers and define parameter values
4. Train the LSTM network with the specified training options by using train Network
5. Feature extraction with activation functions
6. Test the data using SVM classifier.
7. Calculate the classification accuracy of the predictions.
8. Output: accuracy

the operations performed on each data set separation, the maximum average success value obtained using the Resnet50 network architecture was found to be 61 % in the Subspace Discriminant classifier. The maximum average success value obtained using the DenseNet-201 network architecture was found to be 56.48 % in the Medium Neural Network classifier. The maximum average success value obtained using the Xception network architecture was found to be 61.58 % in the Large Neural Network classifier.

In this study, classification studies were carried out for 13 labels on a total of 72,660 data using different architectures. All the results obtained regarding the data set are given in Table 9.

The maximum classification success of the study in which the data set was obtained is 84.70 %.

A high success value could not be obtained in the classification processes performed with the RGB images in the data set. The classification success obtained with the raw data set was higher than the success values obtained with RGB images. The success value obtained was tried to be improved by using BFS. However, no extreme improvement in classification accuracy was observed with the BFS method. In this study, it was observed that this success value increased in the feature extraction method with LSTM. When the results obtained in this study are compared with the literature, it seems that there is a significant improvement compared to the literature. Studies that do not use the same data set but classify human activities with the kinect sensor were also examined. Although it is not correct to directly compare the successes achieved in these data sets, it will give information about the place of our study in the literature. They used 3 different data sets which have 16 different activities. They achieved 88.64 %, 89 % and 85.70 % accuracies for three different data sets [69]. Saini et al. classified 24 human activities by using kinect sensor and they obtained 68.9 % accuracy in their studies [70]. Ghogog et al. proposed a method for fish-erposes activities by using kinect sensor data. These studies show that as

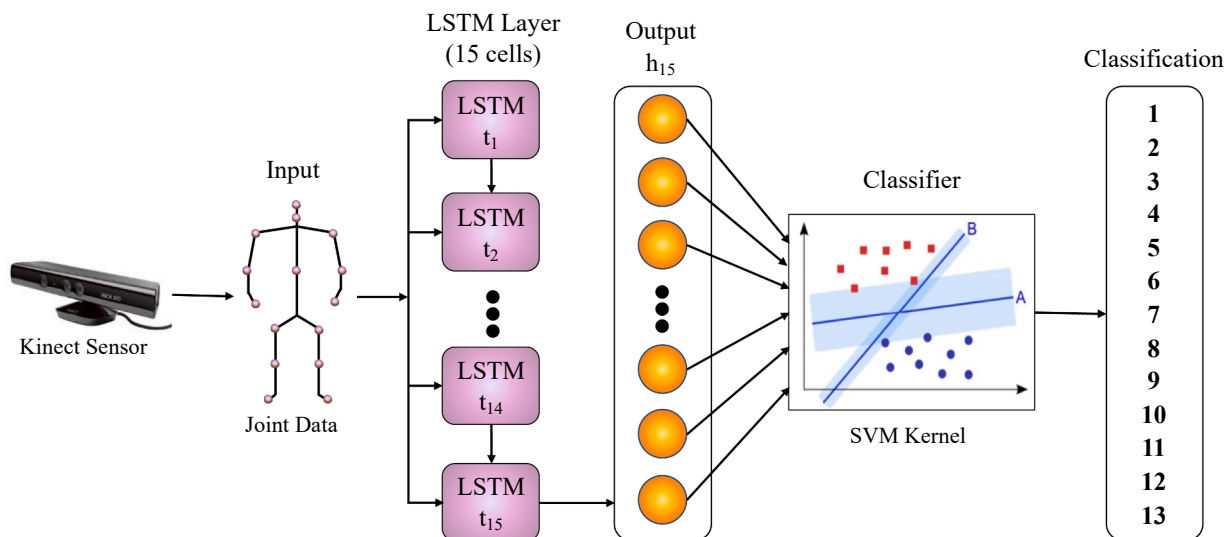


Fig. 8. Framework of feature extraction method with LSTM.

the number of activities increases, the success decreases. For this purpose, the framework of the proposed method to increase the success in this study is shown in Fig. 8. Pseudocode for this method is given in Table 10.

When the framework of the method is examined, firstly, a data set is obtained using the kinect sensor. This data set consists of joint position and orientation information. In the third stage, LSTM architecture is used to perform feature extraction in the dataset. By using the obtained feature matrix, the classification process is carried out by machine learning method. The maximum success value from the machine learning-based classifiers was obtained in the SVM Kernel classifier.

6. Conclusion

In recent years, numerous studies have been carried out to detect the movements of people. This study was carried out with the motivation of detecting the erroneous stances that people take in business life. In this study, classification studies were conducted using machine learning-based and deep learning-based methods for 13 labels on 72,660 data.

When classical machine learning methods were used for the raw sensor data obtained from the joints, the highest success value was obtained as 72.60 % with the Cosine K-Nearest Neighbour classification method. However, higher success was achieved with the Backward Feature Selection method (74.88 %). When feature extraction was done with the Long Short Term Memory architecture, which is one of the deep learning methods for the same data, and machine learning-based classification methods are used, maximum success value was obtained (98.85 %). Classification was made with Learning from scratch (24.24 %) and transfer learning (61.58 %) methods using RGB images. By looking at the classification results made by selecting features for joint data, it is observed that classification success is increased by using features that play a significant role in data classification. The biggest contributor to this increase is used LSTM architecture as a feature extractor. In the classification of human movements, it was found that joint data gave much more successful results than image data.

In future studies, deep learning architectures can be developed with more RGB camera images to ensure distinctiveness with data obtained only from the camera image. Additionally, in order to adjust the ergonomics of the working environment in future studies, it is aimed to design a system that detects the erroneous postures of the employees and warns them when they are in a wrong posture.

CRedit authorship contribution statement

Büşra Açı̇s: Formal analysis, Writing – original draft, Visualization, Methodology, Data curation, Investigation, Software. **Selda Güney:** Conceptualization, Validation, Project administration, Investigation, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] B. Galna, G. Barry, D. Jackson, D. Mhiripiri, P. Olivier, L. Rochester, Accuracy of the Microsoft Kinect sensor for measuring movement in people with Parkinson's disease, *Gait Post.* 39 (2014) 1062–1068, <https://doi.org/10.1016/j.gaitpost.2014.01.008>.
- [2] S. Gasparrini, E. Cipitelli, S. Spinsante, E. Gambi, A Depth-Based Fall Detection System Using a Kinect® Sensor, *Sensors* 14 (2014) 2756–2775, <https://doi.org/10.3390/s140202756>.
- [3] V. Bevilacqua, N. Nuzzolese, D. Barone, M. Pantaleo, M. Suma, D. D'Ambruso, A. Volpe, C. Loconsole, F. Stroppa, Fall detection in indoor environment with kinect sensor, in: 2014 IEEE International Symposium on Innovations in Intelligent Systems and Applications (INISTA) Proceedings, IEEE, 2014: pp. 319–324. <https://doi.org/10.1109/INISTA.2014.6873638>.
- [4] N. Asilah Saidin, S.A. Abdul Shukor, An Analysis of Kinect-Based Human Fall Detection System, *Proceeding - 2020 IEEE 8th Conference on Systems, Process and Control, ICSPC 2020*. (2020) 220–224. <https://doi.org/10.1109/ICSPC50992.2020.9305797>.
- [5] T. Wang, C. Li, C. Wu, C. Zhao, J. Sun, H. Peng, X. Hu, B. Hu, A Gait Assessment Framework for Depression Detection Using Kinect Sensors, *IEEE Sens. J.* 21 (2021) 3260–3270, <https://doi.org/10.1109/JSEN.2020.3022374>.
- [6] B. Yao, H. Gao, X. Su, Human motion recognition by three-view kinect sensors in virtual basketball training, *IEEE Region 10 Annual International Conference, Proceedings/TENCON*. (2020-Novem (2020)) 1260–1265, <https://doi.org/10.1109/TENCON50793.2020.9293892>.
- [7] T. Saratean, M. Antal, C. Pop, T. Cioara, I. Anghel, I. Salomie, A Physiotherapy Coaching System based on Kinect Sensor, *Proceedings - 2020 IEEE 16th International Conference on Intelligent Computer Communication and Processing, ICCP 2020*. (2020) 535–540. <https://doi.org/10.1109/ICCP51029.2020.9266178>.
- [8] A.D.Q. Burle, T.B.D.G. Lafayette, J.R. Fonseca, V. Teichrieb, A.E.F. Da Gama, Real-time approach for gait analysis using the Kinect v2 sensor for clinical assessment purpose, *Proceedings - 2020 22nd Symposium on Virtual and Augmented Reality, SVR 2020* (2020) 144–153, <https://doi.org/10.1109/SVR51698.2020.00034>.
- [9] Z. Xu, Y. Zhang, C. Fu, L. Liu, C. Chen, W. Xu, S. Guo, B.S. Measurement, T.-D. Reconstruction, of Spinal Shape Using One Kinect Sensor, in: IEEE 17th International Symposium on Biomedical Imaging (ISBI), IEEE 2020 (2020) 1–5, <https://doi.org/10.1109/ISBI45749.2020.9098481>.
- [10] J. Kondragunta, G. Hirtz, Gait Parameter Estimation of Elderly People using 3D Human Pose Estimation in Early Detection of Dementia, in: *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2020-July (2020), pp. 5798–5801, <https://doi.org/10.1109/EMBC44109.2020.9175766>.
- [11] N.K. Mangal, A.K. Tiwari, Kinect v2 tracked Body Joint Smoothing for Kinematic Analysis in Musculoskeletal Disorders, in: *Proceedings of the Annual international Conference of the IEEE Engineering in Medicine and Biology Society*, 2020-July (2020), pp. 5769–5772, <https://doi.org/10.1109/EMBC44109.2020.9175492>.
- [12] S. Nehra, J.L. Raheja, Unobtrusive and Non-Invasive Human Activity Recognition using Kinect Sensor, *Indo - Taiwan 2nd International Conference on Computing, Analytics and Networks, Indo-Taiwan ICAN 2020 - Proceedings*. (2020) 58–63. <https://doi.org/10.1109/Indo-TaiwanICAN48429.2020.9181359>.
- [13] T.H. Tan, M. Gochoo, H.S. Chen, S.H. Liu, Y.F. Huang, Activity Recognition Based on DCNN and Kinect RGB Images, in: 2020 International Conference on Fuzzy Theory and Its Applications, IFUZZY, 2020. (2020), pp. 60–63, <https://doi.org/10.1109/IFUZZY50310.2020.9297815>.
- [14] T. Kamizono, H. Abe, K. Baba, S. Takano, K. Murakami, Towards activity recognition of learners by kinect, *Proceedings - 2014 IIAI 3rd International Conference on Advanced Applied Informatics, IIAI-AAI 2014*. (2014) 177–180. <https://doi.org/10.1109/IIAI-AAI.2014.45>.
- [15] M. Li, F. Wei, Y. Li, S. Zhang, G. Xu, Three-Dimensional Pose Estimation of Infants Lying Supine Using Data from a Kinect Sensor with Low Training Cost, *IEEE Sens. J.* 21 (2021) 6904–6913, <https://doi.org/10.1109/JSEN.2020.3037121>.
- [16] W. Ren, O. Ma, H. Ji, X. Liu, Human Posture Recognition Using a Hybrid of Fuzzy Logic and Machine Learning Approaches, *IEEE Access* 8 (2020) 135628–135639, <https://doi.org/10.1109/ACCESS.2020.3011697>.
- [17] Ö.F. İnce, İ.F. İnce, M.E. Yıldırım, J.S. Park, J.K. Song, B.W. Yoon, Human activity recognition with analysis of angles between skeletal joints using a RGB-depth sensor, *ETRI J.* 42 (2020) 78–89, <https://doi.org/10.4218/etrij.2018-0577>.
- [18] L.T. De Paolis, V. De Luca, The performance of Kinect in assessing the shoulder joint mobility, *IEEE Medical Measurements and Applications, MeMeA 2020 - Conference Proceedings*. (2020) 1–6, <https://doi.org/10.1109/MeMeA49120.2020.9137213>.
- [19] P.K. Pisharady, M. Saerbeck, Kinect based body posture detection and recognition system, *International Conference on Graphic and Image Processing (ICGIP)* 2012). 8768 (2013) 87687F. <https://doi.org/10.1117/12.2009926>.
- [20] W.J. Wang, J.W. Chang, S.F. Huang, R.J. Wang, Human Posture Recognition Based on Images Captured by the Kinect Sensor, *Int. J. Adv. Rob. Syst.* 13 (2016) 1–16, <https://doi.org/10.5772/62163>.
- [21] A. Abobakr, D. Nahavandi, J. Iskander, M. Hossny, S. Nahavandi, M. Smets, RGB-D human posture analysis for ergonomic studies using deep convolutional neural network, *2017 IEEE International Conference on Systems, Man, and Cybernetics, SMC 2017*. 2017-Janua (2017) 2885–2890. <https://doi.org/10.1109/SMC.2017.8123065>.
- [22] R. Kumarasiri, A. Niroshan, Z. Lantra, T. Madusanka, C.U.S. Edussooriya, R. Rodrigo, Gait Analysis Using RGBD Sensors, in: *International Conference on Control, Automation, Robotics and Vision (ICARCV)*, 2018, pp. 460–465, <https://doi.org/10.1109/ICARCV.2018.8581295>.
- [23] Y. Xu, J. Chen, Q. Yang, Q. Guo, H.P. Recognition, fall detection Using Kinect V2 Camera, in: *Chinese Control Conference (CCC)*, IEEE 2019 (2019) 8488–8493. <https://doi.org/10.23919/ChiCC.2019.8865732>.
- [24] H. Sun, G. Zhu, X. Cui, J.-X. Wang, Kinect-based intelligent monitoring and warning of students' sitting posture, in: *In: 2021 6th International Conference on*

- Automation, Control and Robotics Engineering (CACRE), 2021, pp. 338–342, <https://doi.org/10.1109/CACRE52464.2021.9501372>.
- [25] S.H. Chowdhury, M. Al Amin, A.K.M.M. Rahman, M.A. Amin, A.A. Ali, in: Assessment of Rehabilitation Exercises From Depth Sensor Data, in, 2021, pp. 1–7, <https://doi.org/10.1109/ICIT54785.2021.9689826>.
- [26] D.K. Vishwakarma, K. Jain, Three-dimensional human activity recognition by forming a movement polygon using posture skeletal data from depth sensor, *ETRI J.* 44 (2) (2022) 286–299.
- [27] F. Ahmed, A.S.M.H. Bari, M.L. Gavrilova, Emotion Recognition From Body Movement, *IEEE Access* 8 (2020) 11761–11781, <https://doi.org/10.1109/ACCESS.2019.2963113>.
- [28] C. Jaeyong Sung, B. Ponce, A.S. Selman, Unstructured human activity detection from RGBD images, in: In: 2012 IEEE International Conference on Robotics and Automation, 2012, pp. 842–849, <https://doi.org/10.1109/ICRA.2012.6224591>.
- [29] P.D. Barua, N.F. Muhammad Gowdh, K. Rahmat, N. Ramli, W.L. Ng, W.Y. Chan, M. Kuluozturk, S. Dogan, M. Baygin, O. Yaman, T. Tuncer, T. Wen, K.H. Cheong, U. R. Acharya, Automatic COVID-19 Detection Using Exemplar Hybrid Deep Features with X-ray Images, *Int. J. Environ. Res. Public Health* 18 (2021) 8052, <https://doi.org/10.3390/ijerph18158052>.
- [30] H. Liu, G. Cui, Y. Luo, Y. Guo, L. Zhao, Y. Wang, A. Subasi, S. Dogan, T. Tuncer, Artificial Intelligence-Based Breast Cancer Diagnosis Using Ultrasound Images and Grid-Based Deep Feature Generator, *International Journal of, General Medicine*. 15 (2022) 2271–2282, <https://doi.org/10.2147/IJGM.S347491>.
- [31] P.D. Barua, W.Y. Chan, S. Dogan, M. Baygin, T. Tuncer, E.J. Ciaccio, N. Islam, K. H. Cheong, Z.S. Shahid, U.R. Acharya, Multilevel Deep Feature Generation Framework for Automated Detection of Retinal Abnormalities Using OCT Images, *Entropy* 23 (2021) 1651, <https://doi.org/10.3390/e23121651>.
- [32] E. Aydemir, M.A. Yalcinkaya, P.D. Barua, M. Baygin, O. Faust, S. Dogan, S. Chakraborty, T. Tuncer, U.R. Acharya, Hybrid Deep Feature Generation for Appropriate Face Mask Use Detection, *Int. J. Environ. Res. Public Health* 19 (2022) 1939, <https://doi.org/10.3390/ijerph19041939>.
- [33] X. Xu, R.W. McGorry, The validity of the first and second generation Microsoft Kinect™ for identifying joint center locations during static postures, *Appl. Ergon.* 49 (2015) 47–54, <https://doi.org/10.1016/j.apergo.2015.01.005>.
- [34] L. Cruz, D. Lucio, L. Velho, Kinect and RGBD Images: Challenges and Applications, in: 2012 25th SIBGRAPI Conference on Graphics, Patterns and Images Tutorials, IEEE, 2012: pp. 36–49, <https://doi.org/10.1109/SIBGRAPI-T.2012.13>.
- [35] M. Xu, Z. Lei, J. Yang, Estimating the Dead Space Volume Between a Headform and N95 Filtering Facepiece Respirator Using Microsoft Kinect, *J. Occup. Environ. Hyg.* 12 (2015) 538–546, <https://doi.org/10.1080/15459624.2015.1019078>.
- [36] J. Abbasi, H. Salarieh, A. Alasty, A motion capture algorithm based on inertia-Kinect sensors for lower body elements and step length estimation, *Biomed. Signal Process. Control* 64 (2021), 102290, <https://doi.org/10.1016/j.bspc.2020.102290>.
- [37] Z. Zheng, Q. Wang, D. Deng, Q. Wang, W. Huang, CG-Recognizer: A biosignal-based continuous gesture recognition system, *Biomed. Signal Process. Control* 78 (2022), 103995, <https://doi.org/10.1016/j.bspc.2022.103995>.
- [38] V. Nasteski, An overview of the supervised machine learning methods, *HORIZONS*. B. 4 (2017) 51–62, <https://doi.org/10.20544/HORIZONS.B.04.1.17.P05>.
- [39] Ning Li, Li Zhao, Ai-Xia Chen, Qing-Wu Meng, Guo-Fang Zhang, A new heuristic of the decision tree induction, in: 2009 International Conference on Machine Learning and Cybernetics, IEEE, 2009: pp. 1659–1664, <https://doi.org/10.1109/ICMLC.2009.5212227>.
- [40] J. Peng, N. Peng Zhang, Riedel, Discriminant Learning Analysis, *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)* 38 (2008) 1614–1625, <https://doi.org/10.1109/TSMCB.2008.2002852>.
- [41] C. Cortes, V. Vapnik, Support-vector networks, *Mach. Learn.* 20 (1995) 273–297, <https://doi.org/10.1007/BF00994018>.
- [42] C. C. Prediction of Heart Disease using Different KNN Classifier, in: 2021 5th International Conference on Intelligent Computing and Control Systems (ICICCS), IEEE, 2021: pp. 1186–1194, <https://doi.org/10.1109/ICICCS51141.2021.9432178>.
- [43] J. Wang, S. Luo, Exploiting Ensemble Method in Semi-Supervised Learning, in: 2006 International Conference on Machine Learning and Cybernetics, IEEE, 2006: pp. 1104–1107, <https://doi.org/10.1109/ICMLC.2006.258568>.
- [44] J. Cai, J. Luo, S. Wang, S. Yang, Feature selection in machine learning: A new perspective, *Neurocomputing* 300 (2018) 70–79, <https://doi.org/10.1016/j.neucom.2017.11.077>.
- [45] S. Khalid, T. Khalil, S. Nasreen, A survey of feature selection and feature extraction techniques in machine learning, in, Science and Information Conference, IEEE 2014 (2014) 372–378, <https://doi.org/10.1109/SAL.2014.6918213>.
- [46] K.A. Taher, B. Mohammed Yasin Jisan, M.M. Rahman, Network Intrusion Detection using Supervised Machine Learning Technique with Feature Selection, in: 2019 International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST), IEEE, 2019: pp. 643–646, <https://doi.org/10.1109/ICREST.2019.8644161>.
- [47] S. Mustafa, Feature selection using sequential backward method in melanoma recognition, in: 2017 13th International Conference on Electronics, Computer and Computation (ICECCO), IEEE, 2017: pp. 1–4, <https://doi.org/10.1109/ICECCO.2017.8333341>.
- [48] S. Visalakshi, V. Radha, A literature review of feature selection techniques and applications: Review of feature selection in data mining, in: 2014 IEEE International Conference on Computational Intelligence and Computing Research, IEEE, 2014: pp. 1–6, <https://doi.org/10.1109/ICCIC.2014.7238499>.
- [49] G.B. Ergün, S. Güney, A Comparison of the Multivariate Calibration Methods with Feature Selection for Gas Sensors ' Long - Term Drift Effect, *International Journal of Technological Sciences*. 11 (2019) 170–176.
- [50] K.R. Swetha, N. M. A.M. P. M.Y. M, Prediction of Pneumonia Using Big Data, Deep Learning and Machine Learning Techniques, in: 2021 6th International Conference on Communication and Electronics Systems (ICCES), IEEE, 2021: pp. 1697–1700, <https://doi.org/10.1109/ICCES51350.2021.9489188>.
- [51] M. Dol, A. Geetha, A Learning Transition from Machine Learning to Deep Learning: A Survey, in: 2021 International Conference on Emerging Techniques in Computational Intelligence (ICETCI), IEEE, 2021: pp. 89–94, <https://doi.org/10.1109/ICETCI51973.2021.9574066>.
- [52] Y. Lecun, L. Bottou, Y. Bengio, P. Haffner, Gradient-based learning applied to document recognition, *Proceedings of the IEEE*. 86 (1998) 2278–2324, <https://doi.org/10.1109/5.726791>.
- [53] H. Srivastava, K. Sarawadekar, A Depthwise Separable Convolution Architecture for CNN Accelerator, in: 2020 IEEE Applied Signal Processing Conference (ASPCON), IEEE, 2020: pp. 1–5, <https://doi.org/10.1109/ASPCON49795.2020.9276672>.
- [54] Y. Heryadi, H.L.H.S. Warnars, Learning temporal representation of transaction amount for fraudulent transaction recognition using CNN, Stacked LSTM, and CNN-LSTM, in: 2017 IEEE International Conference on Cybernetics and Computational Intelligence (CyberneticsCom), IEEE, 2017: pp. 84–89, <https://doi.org/10.1109/CYBERNETICSCOM.2017.8311689>.
- [55] G.B. Ergün, S. Güney, T.G. Ergün, Classification of Long Bones in Dogs Using Convolutional Neural Networks, *Firat University Journal of, Eng. Sci.* 33 (2021) 125–132, <https://doi.org/10.35234/fumdb.759340>.
- [56] J. Gu, M. Zhang, T. Chen, Z. Sun, Y. Ma, B. Yu, Recent advances in convolutional neural network acceleration, *Pattern Recognit.* 77 (2018) 354–377, <https://doi.org/10.1016/j.neucom.2018.09.038>.
- [57] M.A. Ozdemir, D.H. Kisa, O. Guren, A. Akan, Hand gesture classification using time-frequency images and transfer learning based on CNN, *Biomed. Signal Process. Control* 77 (2022), 103787, <https://doi.org/10.1016/j.bspc.2022.103787>.
- [58] G.B. Ergun, S. Güney, Classification of Canine Maturity and Bone Fracture Time Based on X-Ray Images of Long Bones, *IEEE Access* 9 (2021) 109004–109011, <https://doi.org/10.1109/ACCESS.2021.3101040>.
- [59] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, *Proceedings - 30th IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017*. 2017-Janua (2017) 2261–2269, <https://doi.org/10.1109/CVPR.2017.243>.
- [60] F. Chollet, Xception: Deep Learning with Depthwise Separable Convolutions, in: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, 2017: pp. 1800–1807, <https://doi.org/10.1109/CVPR.2017.195>.
- [61] K. He, X. Zhang, S. Ren, J. Sun, Deep Residual Learning for Image Recognition, in: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, 2016: pp. 770–778, <https://doi.org/10.1109/CVPR.2016.90>.
- [62] S. Hochreiter, J. Schmidhuber, Long Short-Term Memory, *Neural Comput.* 9 (1997) 1735–1780, <https://doi.org/10.1162/neco.1997.9.8.1735>.
- [63] Q. Ye, X. Yang, C. Chen, J. Wang, River Water Quality Parameters Prediction Method Based on LSTM-RNN Model, in: Chinese Control And Decision Conference (CCDC), IEEE 2019 (2019) 3024–3028, <https://doi.org/10.1109/CCDC.2019.8832885>.
- [64] E. Dougherty, J. Hua, C. Sima, Performance of Feature Selection Methods, *Curr. Genomics* 10 (2009) 365–374, <https://doi.org/10.2174/138920209789177629>.
- [65] C. Sima, E.R. Dougherty, The peaking phenomenon in the presence of feature-selection, *Pattern Recogn. Lett.* 29 (2008) 1667–1674, <https://doi.org/10.1016/j.patrec.2008.04.010>.
- [66] R. Ghanbari, K. Borna, M.-S. Using, L.S.T.M. Neural Networks, in, 26th International Computer Conference, Computer Society of Iran (CSICC), IEEE 2021 (2021) 1–5, <https://doi.org/10.1109/CSICC52343.2021.9420543>.
- [67] J. Wang, W. Wang, S. Wei, Y. Zeng, F. Luo, Time Series Sequences Classification with Inception and LSTM Module, in: 2019 IEEE International Conference on Integrated Circuits, Technologies and Applications (ICTA), IEEE, 2019: pp. 51–55, <https://doi.org/10.1109/ICTA48799.2019.9012862>.
- [68] L. Hosseini, R. Shaghghi Kandovan, Hyperspectral Image Classification Based on Hierarchical SVM Algorithm for Improving Overall Accuracy, *Advances in Remote Sensing*. 06 (2017) 66–75, <https://doi.org/10.4236/ars.2017.61005>.
- [69] B. Ghogh, H. Mohammadzade, M. Mokari, Fisherposes for Human Action Recognition Using Kinect Sensor Data, *IEEE Sens. J.* 18 (2018) 1612–1627, <https://doi.org/10.1109/JSEN.2017.2784425>.
- [70] R. Saini, P. Kumar, P.P. Roy, D.P. Dogra, A novel framework of continuous human-activity recognition using Kinect, *Neurocomputing* 311 (2018) 99–111, <https://doi.org/10.1016/j.neucom.2018.05.042>.